

The political connection

President Obama's inauguration was broadcast using Silverlight

The Olympic connection

NBC used Silverlight to broadcast the 2008 Summer Olympic games and the 2010 Winter Olympic games

Support for mouse wheel

In previous versions of Silverlight, you had to rely on some helper classes from either DeepZoom or other sample sites to implement handling the mouse's scroll wheel functionality on things like ListBox, etc. But with Silverlight 4, your applications can respond to mouse wheel events on controls where scrolling is appropriate. The Silverlight ScrollViewer automatically handles the MouseWheel event so the ListBox, ComboBox, TextBox, and ScrollViewers automatically scroll. The best part about the MouseWheel event is that it exists on all UIElements. This event fires whenever the mouse wheel is turned while the mouse pointer is over a UIElement. You can code the MouseWheel event in your application with a code that looks like:

```
private void
OnMouseWheel(object sender,
MouseWheelEventArgs e)
{
    if (e.Delta > 0)
        slider_X.Value += slider_X.
LargeChange;
    else
        slider_X.Value -= slider_X.
LargeChange;
}
```

Support for web cams and microphones

Flash had it for ages. But finally, Silverlight 4 comes with web cam and microphone support. It uses the device capture API to determine which video and audio devices are connected to the computer. The same API is then used to retrieve the devices, prompt the user for permission to access the devices, and use to the devices to capture audio and/or video.

Now once the user has granted permission, you can use the CaptureDeviceConfiguration class' GetDefaultVideoCaptureDevice method to get the default video device and the GetDefaultAudioCaptureDevice method to get the default audio device. This video can be captured by creating a new instance of the CaptureSource class and setting its VideoCaptureDevice property to the video device. Let's see the code necessary to request permission to your users to



Grant it permission by clicking on Yes

leverage their capture devices and then capture both the audio and video.

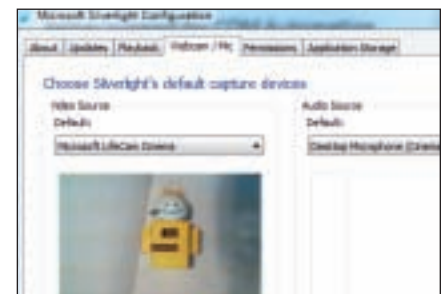
```
We request for permission with:
if (CaptureDeviceConfiguration.AllowedDeviceAccess || Cap-
tureDeviceConfiguration.Request-
DeviceAccess())
{
    _captureSource.Start();
}
Then capture the video using:
if (_captureSource != null)
{
    _captureSource.Stop(); // stop
whatever device may be capturing
_captureSource.VideoCaptureDe-
vice = (VideoCaptureDevice)Vide-
oSources.SelectedItem;
_captureSource.AudioCapture-
Device = (AudioCaptureDevice)
AudioSources.SelectedItem;
VideoBrush vidBrush = new
VideoBrush();
vidBrush.SetSource(_capture-
Source);
if (CaptureDeviceConfiguration.AllowedDeviceAccess || Cap-
tureDeviceConfiguration.Request-
DeviceAccess())
{
    _captureSource.Start();
}
}
In case you just want to take a snap-
shot from the webcam, you need to use:
private void TakeSnapshot_
Click(object sender, RoutedEventArgs e)
{
    if (_captureSource != null)
    {
        _captureSource.
AsyncCaptureImage((snapImage) =>
{
    _images.Add(snapImage);
});
}
}
```

Pretty neat, right? Of course the user first needs to set the default video and audio devices for the Silverlight application by right-clicking on the Silverlight application and going to the web cam/Mic tab as shown below.

Like good old Flash, you can even view and edit the permissions granted for the webcam and microphone access from the configurations window.

Elevated trust applications

Silverlight 3's out-of-browser (OOB) applications had very limited privileges. But with Silverlight 4, you can now



Make the necessary changes in your default devices

alter the OOB manifest to request more elevated permissions for your application. These applications then require the user to grant consent at install time and run outside the browser with elevated trust. To enable a trusted application,



Click on the highlighted check box

Open the project in Visual Studio and check the "Require elevated trust when running outside the browser" permissions checkbox in the Out-of-Browser settings for the Silverlight project. The result will be that the OOB install dialog now looks a little different and warns the user of these elevated privileges.

Local File System Access

In order to read/write data to the user's machine, you normally had to do it